CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2016 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Which device is used for Modulation and Demodulation? Explain its function.**

Answer: A **MODEM (Modulator-Demodulator)**. Converts digital computer pulses to analog signals for transmission over analog lines (modulation) and translates received analog waves back into digital bits (demodulation).

Q2 (3M) **What is Network Topology? List advantages and disadvantages of Mesh Topology.**

Mesh Topology:

- *Advantages:* Dedicated point-to-point links eliminate traffic congestion; robust fault tolerance (link failure affects only two nodes); excellent privacy.
- *Disadvantages:* Requires massive cabling (N(N-1)/2 links) and expensive multi-port hardware; complex installation.

Q3 (3M) **Differentiate between Simplex, Half-Duplex, and Full-Duplex Transmission.**

Differences: Simplex is unidirectional, Half-duplex is alternate bidirectional, Full-duplex is simultaneous bidirectional.

Q4 (3M) **What is Bandwidth of a channel?**

Definition: The range of frequencies contained in a signal or passed by a transmission channel: B = f_{max} - f_{min} (measured in Hertz, or bps in digital communication).

Q5 (3M) **What is a Protocol in Computer Networks?**

Definition: A formal set of rules and conventions governing the format, timing, sequencing, and error control of data transmission between communicating entities (Syntax, Semantics, Timing).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain Seven Layers of ISO/OSI Reference Model.**

OSI Model: Physical, Data Link, Network, Transport, Session, Presentation, Application layers with primary duties.

Q7 (7.5M) **Explain Sliding Window Protocol: Stop-and-Wait vs Sliding Window.**

Comparison: Stop-and-Wait sends 1 frame at a time; Sliding Window allows sending W frames before receiving acknowledgments, multiplying channel utilization.

Q8 (7.5M) **Explain Distance Vector Routing Algorithm with Bellman-Ford equation.**

Algorithm: Path updating using D_x(y) = \min_v \{ c(x, v) + D_v(y) \} across neighboring nodes.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain Error Control using CRC with an example bit sequence.**

CRC Division: Modulo-2 binary arithmetic and frame checksum verification.

Q10 (15M) **Explain Guided Media: Twisted Pair, Coaxial Cable, and Optical Fiber.**

Guided Media: Full analysis of transmission mechanisms, attenuation, and physical connectors.

Q11 (15M) **Explain Switching Techniques: Circuit Switching, Message Switching, and Packet Switching.**

Switching Methods: Architectural breakdown, delay calculations, and applications.

Q12 (15M) **Explain Congestion Control: Leaky Bucket and Token Bucket Algorithms.**

Traffic Shaping: Detailed algorithms, diagrams, and burst handling.

Q13 (15M) **Explain IP Addressing (IPv4 Classes A, B, C, D, E) and Subnetting.**

IP Addressing: Address format, network IDs, host IDs, default masks, and subnet mask calculation.
